

Using NLP to Analyze the Role of Socioeconomic Factors in Transportation & Health Outcomes

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Abstract: Urban transportation systems significantly affects health outcomes, with high traffic exposure resulting in cardiovascular and respiratory diseases and elevated levels of stress. This study aims to utilize natural language processing to understand how transportation is framed in relation to public health. This study used Latent Dirichlet Allocation (LDA) for topic modelling, SBERT and Vader for sentiment analysis, KMeans Clustering for thematic categorization, and n-gram analysis for understanding linguistic patterns. It was found that negative sentiment was strongly associated with traffic exposure, socioeconomic disparities, and urban neglect. Descriptive analyses tended to focus more on mental health stressors than empirical analyses, which focused more on physical health. Additionally, n-gram analysis confirmed that exposure to traffic is one reason transportation is often seen as a hindrance to society, citing mentions to pollution, decay, and social inequality. To better promote a safe and positive view of transportation, cities should consider a multidisciplinary approach in policy making, considering both physical conditions and psychological concerns people may tackle.

Keywords: Urban Transportation, Public Health, Traffic Exposure, Natural Language Processing (NLP), Sentiment Analysis, Latent Dirichlet Allocation (LDA), KMeans Clustering, NGrams, Socioeconomic Disparities, Urban Planning

1 Introduction

In the United States, about 3.6 million Americans are unable to access medical care (non-emergency appointments) each year due to a lack of reliable access to transportation. Insufficiency of this vital resource for daily life results in discrepancies in access to healthcare, employment opportunities, and physical and mental health. In particular, this disparity disproportionately affects neighborhoods with significant proportions of people of color. People of color have also been found to rely more heavily on public transport compared to those who are white (Wallace et al. 2005).

Other demographic groups are also affected by the disparities of public transportation, limiting their access to essential services and maintain a convenient and efficient life. For example, in Maryland, an association was found between being a single parent with an increased use of public transit compared to the state average. Their strong dependence on public transportation comes from everyday necessities, such as taking their children to school or childcare, buying groceries, and commuting to work. Without access to childcare services, a parent is faced with the predicament of potentially making a life-changing employment decision, or being unable to provide their child with a good education without money for public transit (Wang and Xu 2020).

Beyond daily necessities, limited transportation also affects healthcare. Many individuals, such as elderly adults and single parents, struggle to reach their medical appointments or travel to pharmacies. This leads to delayed treatment and worsening of health conditions. The insufficiency of efficient mobility can exacerbate existing disparities in health, particularly for low-income groups (Mirza and Hulko 2022). Transportation is largely known as a logistical issue that involves delays and on-times, but it is

paramount to shaping public health. Despite its significant impact on society, transportation may receive inadequate attention in public health research compared to other social determinants of health. This paper explores how transportation accessibility is framed in modern public health literature, hypothesizing that it is more frequently mentioned and often viewed as a positive contributor to health.

1.1 Research Question

Research Question: Are public health research studies more likely to mention accessibility to transportation compared to other topics, and how is this relationship framed?

Purpose: to examine how accessibility to transportation is discussed in relation to public health by using NLP techniques to identify key themes, sentiment, and the frequency of associations in textual data from research studies

Objectives:

1. To use NLP techniques to extract and categorize recurring themes in research studies discussing transportation accessibility and public health
2. To compare how often transportation is mentioned relative to other public health concerns (i.e. employment, education, income & wealth, race, and stress)
3. To analyze whether accessibility is discussed as a positive, negative, or neutral factor in health outcomes
4. To identify patterns in how transportation is framed in relation to health (eg beneficial and improving healthcare access, or as a barrier and worsening disparities in health)

2 Background

The social determinants of health (SDOH) are non-medical factors that can influence public health (WHO).

Key Factors Based on The Social Determinants of Health

- **Neighborhood Conditions:** Social and physical environments; individual health.
- **Working Conditions:** Employment-related factors (the physical aspects of working as well as the work environment; psychosocial aspects; opportunities & resources).
- **Education:** Educational opportunities and the social & psychological factors associated.
- **Income & Wealth:** Socioeconomic status, the environment one grows up in, and the resources they have access to.
- **Race & Racism:** Racial disparities in health & healthcare.
- **Stress:** Psychological stressors (their causes & effects).

3 Methods and Results

3.1 Data Collection and Preprocessing

The dataset for the natural language model was compiled from 50 publicly online research articles on public health, published globally from 2016-2025. These articles encompass diverse geographical regions, with the largest proportion (20 percent) focusing on the United States. The public health research articles compiled primarily discussed transportation and health equity amongst minority groups, the role of environmental exposure in cancer, the impact of socioeconomic status and the social determinants of health on access to healthcare, the relationship between walkability and chronic disease, and the effect COVID-19 had on vulnerable populations. The remaining articles examine public health topics in locations such as rural Canada, the Denver metropolitan area in Colorado, Australia, and various low- and middle-income countries. These articles were managed for organization using Mendeley Reference Manager, and their contents were extracted and stored as .txt files in a designated articles folder.

Using Python's Natural Language Toolkit, I created multiple versions of a preprocessing function that removes stopwords, punctuation, and special characters, allowing for flexible text formatting based on specific output requirements. I also changed all the text to lowercase and made sure to remove 'et' and 'al' since these were common in scientific literature but carried no significant meaning. The preprocessing functions used were:

- *preprocess*: takes in a string and returns a list of strings; useful for when we want every individual word in the sentence to be a string, and for every sublist to represent a sentence.
 - e.g. "The cat ran up the hill" → [['cat', 'ran', 'hill']]
- *preprocess2*: takes in a string and returns a list of strings, where each string represents a sentence; useful for maintaining general sentence structure for sentence sentiment analysis
 - e.g. "Jack and Jill went up the hill to fetch a pail of water. Jack fell down and broke his crown, and Jill came tumbling after." → ['Jack Jill went hill fetch pail water.', 'Jack fell broke crown Jill came tumbling.']
- *preprocess3*: takes in a single word and outputs a single word; more beneficial for cleaning tokens at the word level as opposed to the sentence or full text level. This function also using lemmatization, where each word was reduced to its root form.
 - e.g. 'Corpora' → 'corpus'

3.2 Theme Extraction (NLP Topic Modeling) and Results

Keyword Extraction Using TF-IDF

TF-IDF (Term Frequency - Inverse Document Frequency) is a statistical measure that allows people to determine how significant a word is compared to the rest of the corpus. TF-IDF consists of term frequency (how frequently a term appears in a document) and inverse document frequency (uses logarithm to lessen the effect of very common words, such as 'the', while bringing more weight to rare words that may uniquely define the document) In our corpus, the top five TF-IDF words were:

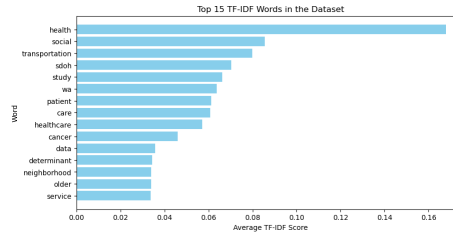


Figure 1: Top 15 TF-IDF Words

Term	TF-IDF Score
health	0.1679
social	0.0855
transportation	0.079
SDOH	0.070
study	0.0661

These words likely suggest that the corpus is likely to focus on topics related to healthcare, social factors, and patient care in the context of transportation and accessibility. ‘Health’ has the highest TF-IDF score, which means that it is a very distinguishing theme across all the documents. Other top-ranking words - ‘social’, ‘transportation’, ‘SDOH’, and ‘study’ - suggest an emphasis on how socioeconomic status, environment, and access to transportation may affect health outcomes (Figure 1).

TF-IDF gives highly common words less weight in order to highlight words particular to the corpus and not general speech, implying that other public health concerns. The absence of words like ‘employment’ or ‘education’ among the top TF-IDF words suggests their usage might not be as distinctive. The prominence of ‘transportation’ reinforces its key role in these discussions.

Earlier, we defined some common themes found in research articles on the social determinants of health: environment, social/psychological, opportunities/resources, and biological effects. Some words are prevalent simply because they come from research. We can sort the top 15 words into common themes as follows, ignoring ‘wa’, which has no substantial importance:

- Biological Effects: ‘health’, ‘patient’, ‘cancer’
- Social/Psychological: ‘social’, ‘SDOH’, ‘determinant’, ‘neighborhood’, ‘older’
- Opportunities/Resources: ‘transportation’, ‘care’, ‘healthcare’, ‘service’
- Research: ‘study’, ‘data’

However, without additional context such as topic modelling, it is difficult to make inferences about the relationships between words. One such way to model topics is through Latent Dirichlet Allocations (LDA).

Latent Dirichlet Allocation (LDA) is a Bayesian network that automatically extracts topics from a corpus. In other words, similar topics are expected to use a similar group of

words. For each document d , the model will draw θ_d , a vector that describes the proportion of each topic in the document. (i.e. how common is each topic in the document?) For each word in the document, the model will select a topic $z_{d,n}$ by drawing from θ_d . (i.e. what topic does the word belong in?) From the topic $z_{d,n}$, the model will choose a word $w_{d,n}$ using ϕ_z , which is a topic-word distribution (multinomial distribution over words for each topic).

The goal is to find the topic distribution for each document ϕ_d (the proportion of topics in each document) and the word distribution for each topic ϕ_k , where k is the number of topics.

Since the hidden topics z are unknown, these parameters are calculated by *maximizing the likelihood* of the observed words in each of the documents using the following equation:

$$p(w|\alpha, \beta) = \prod_{d=1}^D \sum_{\theta_D} \int_{\phi} p(w_d|\theta_d, \phi) * p(\theta_d|\alpha) * p(\phi|\beta) * d\phi,$$

where α is the Dirichlet prior for the topic distribution θ and β is the Dirichlet prior for the word distribution ϕ . A large α means that the document is likely to contain many topics, while a small α means the document is likely to contain fewer. Similarly, small β values mean that the topic should have fewer words compared to topics with greater β values.

In other words, the LDA model assigns each word $w_{d,n}$ a topic and marginalizes (integrates) all possible topic assignments. The Dirichlet distributions for the document-topic and topic-word distributions are also taken into account. These probabilities are multiplied together since there are multiple documents in a corpus and the documents are conditionally independent.

Latent Dirichlet Allocation (LDA) Results: Topic Identification

In this project, LDA was used for classification to identify recurring themes, such as employment, race, or transportation, in relation to health outcomes. An intertopic distance map was generated to visualize all topics in a 2-dimensional space. The size of each bubble on the left is proportional to the number of words included in each topic. By default, the bar chart on the right displays the document's top 30 most salient terms. Saliency and relevance can help to determine the best words that make up a topic.

Saliency: measures the significance of a word in the whole corpus.

$$Saliency(w) = P(w) \sum_k P(k|w) \log \frac{P(k|w)}{P(k)}$$

Relevance: measures how well a word can represent a topic.

$$r(w|k, \lambda) = \lambda P(w|k) + (1 - \lambda) P(w|k) \frac{\log P(w|k)}{\log(P(w))}$$

The four-topic LDA model revealed the most salient terms included 'results', 'block', 'transportation', 'group', and 'traffic', suggesting a focus on transportation and its relationship with public health.

Saliency was used to indicate the most prominent words in the corpus. Since transportation and traffic appeared frequently, they are significant themes in the discussion of social determinants of health. For example, when adjusting the relevance metric λ to 0.7 to balance generality and uniqueness, 3 of the 4 topics contained 'traffic' in their top 5 most relevant terms, suggesting that traffic is a very prevalent theme and issue discussed with transportation.

Latent Dirichlet Allocation (LDA) Topic-Word Distributions

Top 5 words for each topic:

TOPIC 1: ['traffic', 'exposure', 'transportation', 'land', 'processes']

TOPIC 2: ['block', 'group', 'percent', 'percent', 'spatial']
 TOPIC 3: ['traffic', 'exposure', 'high', 'associated', 'higher']
 TOPIC 4: ['results', 'regression', 'college', 'traffic', 'durbin']

- Topic 1: likely emphasizes how **environmental** factors, such as air pollution or noise from traffic, may impact public health
- Topic 2: likely focuses on statistical analysis by using groups and analyzing spatial data. This topic is about **demographics**.
- Topic 3: likely to highlight the association between increased **traffic with negative health outcomes**, such as stress or chronic diseases.
- Topic 4: likely to deal with research and **statistical analysis**. “Durbin” may refer to the Durbin-Watson statistic, which is used in regression analysis.

Latent Dirichlet Allocation (LDA) Interpretation

This model has a coherence score of about -6.711, which suggests poor topic coherence. Coherence is typically in the range of $[-\infty, \infty]$, with higher and positive values indicating that the words in a topic are more semantically meaningful. A low score such as -6.711 may suggest the topics seem random and uninterpretable.

In terms of perplexity, which measures how well a probabilistic model predicts a held-out test set, lower log perplexity scores indicate more accurate predictions. The model’s log perplexity score of -7.000 suggests that the model is relatively confident in its predictions.

Although a low topic coherence suggests the topics are less interpretable, the relevance of terms such as ‘transportation’ and ‘traffic’ point to important themes in the intersection of transportation and health. The low log perplexity score indicates the model is still good for predicting unseen data.

3.3 Sentiment Analysis and Results

After applying LDA to extract the most prevalent topics from the corpus, we have a better understanding of what themes, such as accessibility or public health outcomes, can be revealed within the corpus. Using sentiment analysis, we can then analyse how these themes - especially accessibility to transportation - are framed in a positive, negative, or neutral manner relative to public health.

Using Vader’s `SentimentIntensityAnalyzer()` class, we were able to study the sentiment of text by assigning a numeric polarity score for each sentence. For example, the sentence “Accessible public transportation greatly improves patient access to healthcare services” has a negative score of 0, a neutral score of 0.744, positive score of 0.256, resulting in an overall score of 0.4754. This suggests this sentence is mostly neutral but has some slightly positive sentiment, likely because of the ‘greatly improves’ in the sentence.

I used Sentence Transformers (AKA SBERT) to transform each sentence from my corpus into a vector representation. Using cosine similarity, I was able to keep track of sentences that were more similar to the query, where I searched for ‘transportation accessibility’ and ‘health’. This would provide me with all the accessibility-related sentences from my corpus with a cosine similarity to the query of at least 0.3. In this example, 151 sentences were above the threshold with a mean sentiment score of -0.06.

Since the mean sentiment score of all sentences related to transportation accessibility and health is below 0 (-0.115), on average, there is slight negative sentiment in the text when discussing accessibility. Of the 151 sentences, 17/151 or about 11.26% of the sentences had a sentiment of at least 0.5. 52/151 or about 33.44% of the sentences had a sentiment between -0.5 and 0.5. Lastly, 54/151 or about 35.76% of the sentences had a sentiment less than -0.5. In other words, a small portion of the corpus's sentences were written in a positive light. In fact, two-thirds of the sentences were written either neutrally or in a negative manner.

This brings us to wonder, why is accessibility in relation to health outcomes being discussed so negatively? I stored all the 'negative' sentences in a variable called `negative_sentences` and inspected them. I noticed the following:

- **High Traffic Exposure is a Health Hazard:** high volumes of traffic were presented as an impediment to walking, biking, and using public transportation. People who lived near high volumes of traffic were also more likely to suffer from air pollution, chronic illness, injuries, as well as stress.
- **Socioeconomic Inequalities:** Minority and low-income communities tend to experience more traffic and higher levels of pollution.
- **Perceived Neglect:** Some areas of a city may have a lot of trash and graffiti, which contributes to the city appearing unkempt and unclean, which may discourage people from waiting at certain train or bus stops alone. The perception of an unsafe environment can negatively affect their impression of public transit.

This articles mostly discuss accessibility in a negative light by explaining how traffic and stress are barriers to smooth and efficient transportation and are detrimental to both physical and mental health. There are also socioeconomic inequalities in accessing transportation, and many individuals feel unsafe riding it. However, there is limited research on how transportation affects mental health, particularly in terms of stress in terms of commuting, such as the pressure to get to work or school on time, attend medical appointments, or navigate safety concerns while travelling.

3.4 Comparative Analysis

While sentiment analysis has provided a broad overview of how transportation is discussed positively, negatively, or neutrally in relation to public health, it is important to understand how these sentences matter in the grander scope of health outcomes. I used SBERT with KMeans Clustering to form clusters based on the cosine similarities between all pairs of sentence embeddings tracked in a similarity matrix (Figure 2). I observed two main clusters:

1. **Cluster 0:** Cluster 0 was more theoretical and contextual, covering topics such as abstract transportation systems, urban planning, health disparities, and environmental justice. It included literature reviews on the socioeconomic effects on health. Cluster 0 referenced broader societal issues and tended to be more descriptive. Some examples of Cluster 0 are:
 - “streets are important public spaces but also sources of hazards”
 - “minority populations disproportionately live near high-traffic roads”

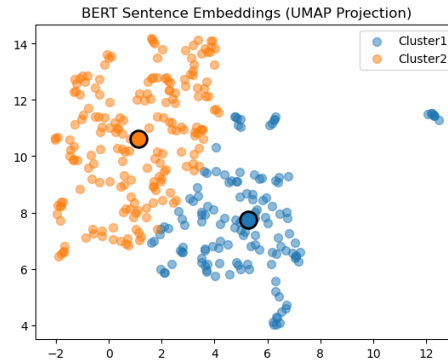


Figure 2: BERT Sentence Embeddings

- “Abstract transportation systems generate certain benefits... yet also source air pollution, noise safety hazards.”
- 2. **Cluster 1:** Cluster 1 focused more on statistical analysis and reported specific findings found with data. Cluster 1 also cites sources of data and highlights local vs regional patterns in disparities in exposure to traffic. It is more quantitative and emphasizes empirical relationships. Some examples from Cluster 1 are:
 - “minority and lower-status populations linked to higher traffic exposure”
 - “poverty is a consistent predictor of traffic density”
 - “Using statistical techniques, we examined variation in residential exposure to traffic.”

After further analysis, Cluster 0 had a predominantly negative tone and focused on problems, such as pollution, health hazards, inequalities, and stress. Although some text had a positive or neutral tone, the text leaned negative. Ex: “High traffic volumes lead to adverse health outcomes... respiratory disease, cardiovascular disease.” On the other hand, Cluster 1 had a neutral to positive sentiment. Its tone was more technical and focused on the hard facts through methods and findings. Ex: “Geographically weighted regression shows variation in traffic exposure” or “Poverty is a consistent predictor of traffic density”.

What accounted for the difference between Clusters 0 and 1? Were certain health outcomes such as mental versus physical health more commonly discussed in one cluster than another?

Cluster 0 discussed both mental and physical health, but emphasized mental health stressors more. Cluster 0 discussed respiratory diseases, cardiovascular diseases, and low birth weight, but also highlighted mental health conditions, including chronic stress, depression, and anxiety from exposure to traffic. In contrast, Cluster 1 almost solely focused on the relationship between traffic and *physical* health. Cluster 0 often critiqued failures in urban planning, while Cluster 1 employed statistics to objectively offer solutions and notice ongoing patterns. Cluster 0, which was more descriptive, also detailed images of physical decay and neglect and linked traffic infrastructure to social disorder, such as in, ‘streets as sites of drug use or crime’. Although it did mention the benefits

of transportation, it is important to note these positives were likely outweighed by the emphasis Cluster 0 provided on identifying existing issues.

While clustering provided insight into thematic distinctions in how transportation is discussed, a broader analysis of frequently occurring phrases can reveal the most pressing concerns in the dataset overall. An analysis using bigrams and trigrams can reveal some of the major issues of public transportation. The top 10 most common trigrams were:

(denver, metropolitan, region), (transportation, land, use), (high, traffic, volumes), (geographically, weighted, regression), (figure, gwr, percent), (neglect, physical, decay), (minority, lower, status), (higher, exposure, traffic), (local, land, use), (air, pollution, noise).

Traffic was heavily discussed and among the most common bigrams, such as in (traffic, exposure), (exposure, traffic), (traffic, density), (high, traffic).

4 Discussion

Using Latent Dirichlet Allocation, we were able to determine some of the recurring themes in research, including the effect of environmental factors on public health, demographic disparities in research studies, the association between increased traffic and negative health outcomes, and statistical analysis.

Relative to other public health concerns, transportation is mentioned more frequently with income and wealth, race, and stress. Sentiment analysis conducted with Vader and SBERT revealed deeper insights into these topics. Negative sentiment was prevalent in discussions about traffic-related health hazards, socioeconomic inequalities, and urban neglect. It is evident that notably, demographics and traffic patterns played a strong role in shaping the general public's perception on public transportation.

The bigram/trigram analysis confirms traffic is indeed a major issue. Phrases like 'High traffic volumes' and other variants with traffic: 'traffic exposure', 'exposure traffic', 'traffic density', and 'high traffic' were frequently present in the corpus bigrams, supporting the notion that negative discussion surrounding accessibility in relation to health outcomes may be due to high traffic exposure. Frequent mentions of the bigrams ('land', 'use') and ('metropolitan', 'region') also suggest an emphasis on urban planning. Socioeconomic disparities were also evident since ('minority', 'lower', 'status') was also a fairly common trigram. Overall, transportation seems to be discussed as a barrier rather than a benefit to society since several phrases were related to the negative impact on environment, such as ('neglect', 'physical', 'decay') and ('air', 'pollution', 'noise').

Interestingly, the results from the SBERT and KMeans Clustering analysis indicate that accessibility is more often discussed as a negative factor in health outcomes in theoretical, descriptive analysis compared to statistical analysis. One reason for this difference may be that empirical analysis has less founded knowledge on physiological well-being compared to emotional health. The sentiment analysis I conducted indicates that transportation accessibility is frequently discussed in relation to physical health concerns, whereas its connection to mental health is less emphasized. This suggests a potential gap in research and policy.

This disparity may stem from less founded research on mental health issues since they are more difficult to quantify and are therefore, taken less seriously compared to complications in physical health. More research is needed to not only build a safer and more efficient transportation network in urban areas, but to bridge the gaps in our

understanding of mental health. Poor mental health is often fueled by insecurity, fear, a lack of awareness. By focusing more on the relationship between transportation and mental health, researchers could take a crucial step towards framing transportation as a benefit rather than a barrier to healthcare access.

Some limitations to note are that I did not sort for a particular region when compiling the research articles in my model. There also may have been some influence in the results from the publishing date/year since I mostly compiled recent articles, but not a particular year. Different publishers from various regions may also be biased to have varying opinions about public transportation, such as writers from urban versus rural locations. Additionally, some articles only had the abstract available, but not the full text. The content of the articles may have also been influenced by the search terms I entered or the website I searched on.

5 Conclusion

This analysis highlights the strong relationship between transportation accessibility and public health, particularly in the context of exposure to traffic, socioeconomic disparities, and urban planning. Sentiment analysis revealed that transportation was often seen as a barrier rather than a benefit. Negative sentiment was often found to be associated with high volumes of traffic, urban neglect, and air pollution. A clustering analysis also revealed a distinction between theoretical critiques of systemic failures and empirical research with statistical analyses, which focused more on physical versus mental health. Since it is less quantifiable, mental health was found to be underexplored in public health research. An approach that integrates urban planning, policy making, and public health would be needed to address the outcomes of both physical and mental health. By implementing new policies, cities can address key concerns such as carbon emissions, traffic exposure, and income disparities. Reframing public transportation can help shift public perception, encouraging people to see it as a benefit rather than a barrier to healthcare access.

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